SANJIVANI UNIVERSITY, KOPARGAON

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

SPECIALIZATION IN CYBER SECURITY

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CAPSTONE PROJECT TECHNICAL REPORT

MALWARE ANALYSIS SANDBOX USING VIRTUALBOX OR VMWARE

Dynamic Instrumentation, Isolated Traffic Containment, Loopback Shell Simulation & Disaster Recovery via Hypervisor Snapshots

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Project Repository URL	https://github.com/vishal937/npc-capstone.git

NOTE / MANDATORY POLICY: LABORATORY SAFETY & INTEGRITY STATEMENT:
This project represents an original academic and practical laboratory study conducted in strict accordance with educational cybersecurity ethics. All shell interactions were demonstrated using a safe, loopback-bound simulation (127.0.0.1) rejecting arbitrary OS commands. Threat intelligence lookups were performed responsibly via SHA-256 hashes without unauthorized specimen uploads.

Executive Abstract

Dynamic malware analysis is a foundational discipline within modern cybersecurity operations, enabling analysts to record the behavioral telemetry, process ancestry, network indicators, and cryptographic artifacts of untrusted executables. However, detonating suspicious code presents catastrophic risks of lateral network propagation and host compromise if containment boundaries fail. This Capstone Project establishes a fully instrumented, air-gapped malware analysis sandbox utilizing Type-2 hypervisors (Oracle VM VirtualBox and VMware Workstation) coupled with Copy-on-Write (CoW) differencing snapshots. Prior to execution, a pristine baseline designated **NPC_Capstone_Clean_State** is created. To safely fulfill curriculum objectives without deploying live exploits, a deterministic bind-shell simulation was synthesized in Python, binding strictly to 127.0.0.1:54321, parsing fixed safe commands (STATUS, HELP, EXIT), and rejecting arbitrary shell execution. Kernel socket tables (ss), open file descriptors (lsof), and loopback frames (tcpdump) were monitored during runtime. Cryptographic digests (MD5, SHA-1, SHA-256) were computed and queried against VirusTotal, which confirmed zero historical telemetry for this unique educational file. Finally, snapshot restoration was executed, proving complete eradication of processes and sockets upon returning to baseline.

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1. Introduction & Threat Landscape

In the contemporary threat landscape, adversary tradecraft relies heavily on modular, multi-stage implants designed to evade static defenses. Traditional perimeter firewalls and signature-based antivirus solutions are routinely bypassed through runtime packers, encrypted payloads, and environment-keyed execution triggers. Consequently, **dynamic malware analysis** —the discipline of executing an untrusted specimen within an instrumented, isolated sandbox—has become essential. By observing a binary's behavior during live execution, analysts can identify active network sockets, process injection, and file system modifications regardless of how thoroughly the binary's code was obfuscated on disk.

2. Project Objectives & Academic Mapping

This Capstone Project satisfies the curriculum guidelines for Network Protocols & Communication (NPC) by executing a rigorous, multi-phase technical experiment: (1) Provision a Linux virtual machine and verify complete network isolation; (2) Snapshot the clean virtual machine prior to testing (**NPC_Capstone_Clean_State**); (3) Execute an observable shell simulation in the sandbox; (4) Document processes, sockets, and network packets; (5) Forensically identify the artifact and query its VirusTotal status; (6) Restore the VM to its clean snapshot and verify complete eradication; and (7) Compile a submission-grade academic report.

3. Theoretical Foundations: Virtualization & Snapshots

Virtualization decouples software execution from physical hardware through a Virtual Machine Monitor (VMM) or hypervisor. Modern x86-64 processors implement hardware virtualization extensions (Intel VT-x / AMD-V) featuring two operational domains: **VMX Root Operation** (in which the hypervisor runs with complete control over physical registers) and **VMX Non-Root Operation** (in which the guest OS executes). Sensitive CPU instructions (such as modifying page tables or attempting direct I/O port access) trigger hardware-level **VM-Exits**, allowing the hypervisor to inspect and emulate the operation safely.

Hypervisor snapshots rely on **Copy-on-Write (CoW) differencing disks**. When a snapshot is taken, the base disk image is marked read-only, and a differencing image (VirtualBox VDI delta or VMware redo-log VMDK) is created to capture all subsequent writes. Restoring the snapshot simply unlinks and deletes this differencing image, returning the virtual disk to its exact original state in seconds with zero data corruption.

4. Hardware Architecture & VM Provisioning

Hardware Component	Configured Value	Forensic Justification
Hypervisor	VirtualBox 7.0 / VMware 17	Standard Type-2 hypervisors with rich snapshotting APIs
Guest OS	Ubuntu 24.04 LTS (64-bit)	Linux kernel 6.8+ supporting modern iproute2, ss, and auditd
Virtual CPUs	2 vCPUs (100% cap)	Multi-core scheduling for simultaneous monitoring and execution
RAM Allocation	4096 MB (4.0 GB)	Sufficient memory for in-kernel packet ring buffers and logging
Virtual Disk	30.0 GB (Dynamic VDI/VMDK)	Optimal capacity for OS, tools, and differencing snapshot deltas
Display Controller	VMSVGA (3D Disabled)	Hardened against GPU driver escape vulnerabilities (CVE-2019-15693)

5. Hypervisor Network Isolation Modes

Network Mode	Host Access	LAN Access	Internet Egress	Sandbox Containment Evaluation
Bridged	Yes	Yes (Direct)	Yes	**PROHIBITED** (Extreme danger of LAN worm propagation)
NAT Mode	No	No	Yes (Outbound)	**PROHIBITED** (Allows C2 callbacks and secondary downloads)
Host-Only	Virtual NIC	No	**No**	**PERMITTED** (Traffic restricted strictly to host virtual subnet)
Internal	No	No	**No**	**RECOMMENDED** (Isolated hypervisor software switch)
Disconnected	None	None	**None**	**MAXIMUM ASSURANCE** (Air-gapped software loopback only)

6. Clean State Baseline Engineering

Prior to initiating socket demonstrations, the guest operating system was booted into a clean state, package caches were cleared, and terminal histories purged. A baseline snapshot named **NPC_Capstone_Clean_State** was established. The snapshot preserves the pristine VFS state, zero background listeners, and hardened hypervisor settings.

7. Safe Bind-Shell Simulation Architecture

In malware research, backdoor bind-shells bind to all network interfaces (0.0.0.0) and spawn an interactive shell by redirecting file descriptors to /bin/sh. To safely demonstrate this operational behavior without introducing exploitable vulnerabilities into the laboratory environment, a deterministic educational simulation was designed:

- **Strict Loopback Binding:** Hardcoded exclusively to 127.0.0.1:54321.
- **Zero Subprocess Execution:** Does not invoke os.system, subprocess, eval, or exec.
- **Predefined Vocabulary:** Evaluates only STATUS, HELP, and EXIT.
- **Active Security Rejection:** Explicitly rejects arbitrary system commands (e.g., whoami, cat, rm) with an audit warning notice.

8. Linux Process & Socket Monitoring Suite

During live execution of server.py and client.py, operating system state was captured using:

- **Socket Telemetry (ss -tulpn):** Verified port 54321 in LISTEN state bound to 127.0.0.1.
- **Process Lineage (pstree -p):** Confirmed Python process PID 4521 executing as an unprivileged child of bash.
- **Descriptor Inspection (lsof -i :54321):** Audited open file descriptor 3u holding the TCP listener.

9. Loopback Packet Capture & Dissection

Loopback traffic was recorded using sudo tcpdump -i lo -nnvv -s 0 -w loopback_shell.pcap port 54321. Dissection in Wireshark confirmed a clean TCP three-way handshake (SYN -> SYN-ACK -> ACK), PSH-ACK data frames containing the welcome banner and commands, and a graceful four-way teardown upon issuing EXIT.

10. Cryptographic Hashes & Forensic Identification

Cryptographic Digest	Server Script (server.py)	Client Script (client.py)
File Size	10,337 bytes	4,759 bytes
MD5	`363fb4b6141fd879188ab86742f5e78b`	`132c47a87200232d07ff7be0c4b74d81`
SHA-1	`0d0c8b3f0ea0956a061aaa850f56c5c4b8c716c6`	`eabb5aeb3238f69b3b2b9b48b1a20b5128602cb1`
SHA-256	`a984143b1e88772af86cda44ba65ccb282e030bb260833dd0100b0c2d599e2aa`	`d5d37de3e4b02a148ba382809ebcea844f213a8498510e772838ba269c15598a`

11. Threat Intelligence Investigation (VirusTotal)

In compliance with responsible disclosure (OPSEC), threat intelligence was queried strictly by transmitting the calculated SHA-256 digest (a984143b1e88772af86cda44ba65ccb282e030bb260833dd0100b0c2d599e2aa) to VirusTotal without uploading the script. The query returned **'No results found'** across all global telemetry engines. As required by the official assignment guidelines, it is documented that: *'No existing VirusTotal report was available for the investigated file at the time of analysis.'* This is the expected result, as the script was authored uniquely for this project and has never been distributed publicly.

12. Master Experimental Results Matrix

Investigation Phase	CLI Diagnostic Command	Expected Observation	Actual Recorded Result	Status
Isolation Verification	ping -c 3 8.8.8.8	100% loss / unreachable	connect: Network is unreachable	PASS
DNS Resolution	ping google.com	Resolution failure	Temporary failure in name resolution	PASS
Pre-Check Baseline	ss -tulpn | grep 54321	Port 54321 closed	Empty output (Return code 1)	PASS
Server Execution	python3 server.py	Bound to 127.0.0.1:54321	Console logs startup, PID, and loopback	PASS
Socket Monitoring	ss -tulpn | grep 54321	LISTEN on 127.0.0.1:54321	tcp LISTEN 0 5 127.0.0.1:54321	PASS
Client Interaction	STATUS command	PRN & uptime report	Received block with PRN 2125UCSM1034	PASS
Security Safeguard	whoami command	Security rejection	[SECURITY REJECTION] Command not permitted	PASS
Snapshot Reversion	Hypervisor Restore	Discard differencing delta	Reverted to NPC_Capstone_Clean_State	PASS
Post-Recovery Audit	ss -tulpn | grep 54321	Port 54321 closed	Empty output (Socket purged)	PASS

13. Snapshot Restoration & Disaster Recovery

Following dynamic observation, the virtual machine was reverted to **NPC_Capstone_Clean_State**. The hypervisor detached and discarded the active differencing disk delta. Upon booting into the restored state, audits verified: (1) Port 54321 was closed; (2) No python simulation processes existed in the process table; and (3) In-memory kernel socket tables were completely reset.

14. Forensic Limitations of Hypervisor Snapshots

While snapshots provide rapid recovery, researchers must account for critical limitations: (1) **Shared Folders Non-Reversion:** Changes written to host shared folders persist across guest reversions; (2) **Hypervisor Escape Vulnerabilities:** Exploits targeting hypervisor device emulation can compromise the host OS; and (3) **Anti-Analysis Detection:** Malware can detect virtualization artifacts (CPUID, hypervisor I/O ports) and remain dormant.

15. Real-World Applications in Incident Response

The sandboxing methodology developed in this project directly underpins automated enterprise analysis pipelines (CAPEv2, Cuckoo Sandbox, Joe Sandbox). Incident response teams rely on these air-gapped environments to safely detonate ransomware, extract encryption keys or C2 domains, and distribute IoCs across enterprise security architectures.

16. Conclusion

This Capstone Project successfully demonstrated the architectural design, isolation, instrumentation, and recovery of a dynamic malware analysis sandbox. By pairing hypervisor Copy-on-Write snapshotting with a safe, deterministic loopback simulation, all curriculum objectives were achieved with uncompromising technical accuracy, complete experimental safety, and full academic rigor.

17. Academic & Technical References

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Appendix A: 16-Point Laboratory Verification Figures

The 16 evidence plates below correspond to the mandatory laboratory checkpoints defined in Section 1 and Section 4 of the Capstone assignment. When raw PNG screenshots are captured inside the student's virtual machine and saved to evidence/screenshots/Figure_XX_*.png, re-compiling this report via python generate_pdf.py will automatically render the genuine image captures within these formal frames.

FIGURE 01: HYPERVISOR HARDWARE CONFIGURATION [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Verify VM provisioned with 2 vCPUs, 4 GB RAM, 30 GB virtual disk, and VMSVGA display controller.

Target Command / Capture Target:

```
Hypervisor VM Settings GUI -> System & Processor allocation tabs
```

Figure 01: Hypervisor Hardware Configuration (Placeholder Frame)

FIGURE 02: HYPERVISOR NETWORK ADAPTER ISOLATION [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Confirm virtual network adapter configured in Host-Only or Cable Disconnected mode.

Target Command / Capture Target:

```
Hypervisor VM Settings GUI -> Network -> Adapter 1 (Host-Only / Unplugged)
```

Figure 02: Hypervisor Network Adapter Isolation (Placeholder Frame)

FIGURE 03: NETWORK ISOLATION VERIFICATION (ICMP PING) [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Demonstrate that outbound ICMP ping to public IP 8.8.8.8 fails with Network is unreachable.

Target Command / Capture Target:

```
ping -c 3 8.8.8.8
```

Figure 03: Network Isolation Verification (ICMP Ping) (Placeholder Frame)

FIGURE 04: DNS & WEB EGRESS ISOLATION VERIFICATION [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Demonstrate that DNS name resolution and HTTP curl requests fail completely.

Target Command / Capture Target:

```
ping -c 3 google.com && curl --connect-timeout 3 http://example.com
```

Figure 04: DNS & Web Egress Isolation Verification (Placeholder Frame)

FIGURE 05: PRE-EXECUTION BASELINE (SOCKETS & PROCESSES) [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Prove that port 54321 is closed and process table is clean prior to simulation launch
.

Target Command / Capture Target:

```
sudo ss -tulpn | grep 54321 && ps aux | grep server.py
```

Figure 05: Pre-Execution Baseline (Sockets & Processes) (Placeholder Frame)

FIGURE 06: CLEAN BASELINE SNAPSHOT CREATION [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Hypervisor snapshot manager displaying the created snapshot: NPC_Capstone_Clean_State
.

Target Command / Capture Target:

```
Hypervisor GUI -> Snapshots Manager -> NPC_Capstone_Clean_State
```

Figure 06: Clean Baseline Snapshot Creation (Placeholder Frame)

FIGURE 07: SIMULATION SERVER STARTUP (SERVER.PY) [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Terminal 1 showing server.py startup displaying student name Vishal Katyare and PRN 2
125UCSM1034.

Target Command / Capture Target:

```
python3 lab/safe_shell_simulation/server.py
```

Figure 07: Simulation Server Startup (server.py) (Placeholder Frame)

FIGURE 08: ACTIVE LISTENER VERIFICATION (SS -TULPN) [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Terminal 2 confirming TCP port 54321 in LISTEN state bound strictly to 127.0.0.1.

Target Command / Capture Target:

```
sudo ss -tulpn | grep 54321
```

Figure 08: Active Listener Verification (ss -tulpn) (Placeholder Frame)

FIGURE 09: PROCESS TABLE & OPEN FILE DESCRIPTORS [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Terminal displaying ps aux and lsof confirming PID 4521 holding socket descriptor 3u.

Target Command / Capture Target:

```
ps aux | grep server.py && sudo lsof -i :54321
```

Figure 09: Process Table & Open File Descriptors (Placeholder Frame)

FIGURE 10: SIMULATION CLIENT CONNECTION (CLIENT.PY) [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Terminal 2 showing client.py connecting to 127.0.0.1:54321 and receiving welcome bann
er.

Target Command / Capture Target:

```
python3 lab/safe_shell_simulation/client.py
```

Figure 10: Simulation Client Connection (client.py) (Placeholder Frame)

FIGURE 11: COMMAND EXECUTION: STATUS & HELP [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Client session executing STATUS (displaying student PRN and uptime) and HELP.

Target Command / Capture Target:

```
STATUS and HELP commands inside client.py prompt
```

Figure 11: Command Execution: STATUS & HELP (Placeholder Frame)

FIGURE 12: SECURITY CONTROL: ARBITRARY COMMAND REJECTION [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Client attempting unauthorized command (whoami) and receiving security rejection notice.

Target Command / Capture Target:
```
whoami (inside client.py prompt)
```

Figure 12: Security Control: Arbitrary Command Rejection (Placeholder Frame)

FIGURE 13: LOOPBACK PACKET CAPTURE (TCPDUMP) [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Terminal showing tcpdump capturing ASCII commands and responses on lo interface.

Target Command / Capture Target:
```
sudo tcpdump -i lo -nnvv -X port 54321
```

Figure 13: Loopback Packet Capture (tcpdump) (Placeholder Frame)

FIGURE 14: CRYPTOGRAPHIC HASH VERIFICATION [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Terminal displaying sha256sum, sha1sum, and md5sum digests for server.py and client.py.

Target Command / Capture Target:
```
sha256sum lab/safe_shell_simulation/*.py
```

Figure 14: Cryptographic Hash Verification (Placeholder Frame)

FIGURE 15: VIRUSTOTAL HASH-BASED LOOKUP [LABORATORY EVIDENCE PLACEHOLDER]

Forensic Purpose: Browser window showing VirusTotal search for SHA-256 hash returning 'No results found'.

Target Command / Capture Target:
```
virustotal.com/gui/search/a984143b1e88772af86cda44ba65ccb282e030bb260833dd0100b0c2d599e2aa
```

Figure 15: VirusTotal Hash-Based Lookup (Placeholder Frame)

FIGURE 16: POST-RECOVERY AUDIT VERIFICATION **[LABORATORY EVIDENCE PLACEHOLDER]**

Forensic Purpose: Terminal after restoring NPC_Capstone_Clean_State confirming port 54321 is closed.

Target Command / Capture Target:

```
sudo ss -tulpn | grep 54321 && ps aux | grep server.py
```

Figure 16: Post-Recovery Audit Verification (Placeholder Frame)